Cyclist Analytical Report

by Corbin Dye

Business Task

Analyze historical bike trip data in order to understand the behavioural differences between annual members and casual users of Cyclistic's services. The resulting analysis insights will be utilized in order to determine what drives users to become annual members and how the marketing team can use digital media to convert casual users to annual members.

Data Sources

In order to complete the business task, bike trip data collected in Chicago over the entire year of 2025 will be used. This public data has been made available by Motivate International Inc under this licence. All riders' personally identifiable information (PII) has been removed from the dataset to prevent data-privacy issues, this will limit the ability to determine exact casual riders that live in the Chicago area.
The provided data is still:

- Reliable: as it is collected by Motivate International Inc. which is the real world operator of Chicago's Divvy bike system.
- Original: collected by Motivated International Inc which makes it first-party data, collected by the sensors in the bikes and by the docking stations.
- Current: the data was collected over the past year (2025)
- Comprehensive: the data collected had over 5.5 million records, which made up essentially the entire population of Divvy bike trips over the last year, and several data points were collected including member types, geo-locations, times, etc.
- Cited: The data was made available under this licence agreement and all PII information was removed.

The data used would help answer our question on how we could increase conversion rates of riders from casuals to members by understanding the differences that exist between the two groups and what approach we could take to target the correct group with our digital media campaign.

Data Processing

Tool Used

- Since the data had more than 5.5 million entries SQL (SQL Server) was the tool of choice to process and analyze this large dataset, as the programme is extremely efficient when working with large datasets.

Data Importing and Consolidation

- The dataset initially consisted of 12 separate tables for each month of the year, these tables were imported into SQL Server and combined into one table (trip_data_full_year_raw) without any alterations as the data and columns for all 12 tables were consistent.

Data Cleaning

The data was then scanned through and cleaned, the cleaning steps involved:

- Removing any entries where the start_time variable was at a later time than the end_time variable.
- Setting all NULL values for start_station and end_station, where bike_type = ‘electric” to ‘On-street’ as these bikes do not have to left at a station as they have GPS trackers, and would negatively affect our analysis if we left these values as NULL or completed removed them.
- Removed all trips that lasted over 24 hours and bike_type = ‘classic’ as it appeared that the system capped rides to 25 hours and this meant that these bikes had not been returned and were most likely lost or stolen.
- Removed all bikes with no geo-location values but almost all these bikes were already categorized as stolen/lost.
- Add all these changes to table new table trip_data_full_year_cleaned.

Data Manipulation

- Created variables
    - ride_length_m: Calculated as the difference in start_time and end_time in minutes.
    - day_of_week: Extracted the day of the week from start_time to determine what day of the week trips were taking place in order to understand the behaviours of users.
    - hour_of_day: Extracted the hour of the day from start_time to determine what hour of the day trips were taking place in order to understand the behaviours of users.
- All data with new manipulations we saved to new table trip_data_analysis.

Analysis and Key Findings

Trip Purpose

Analysis:

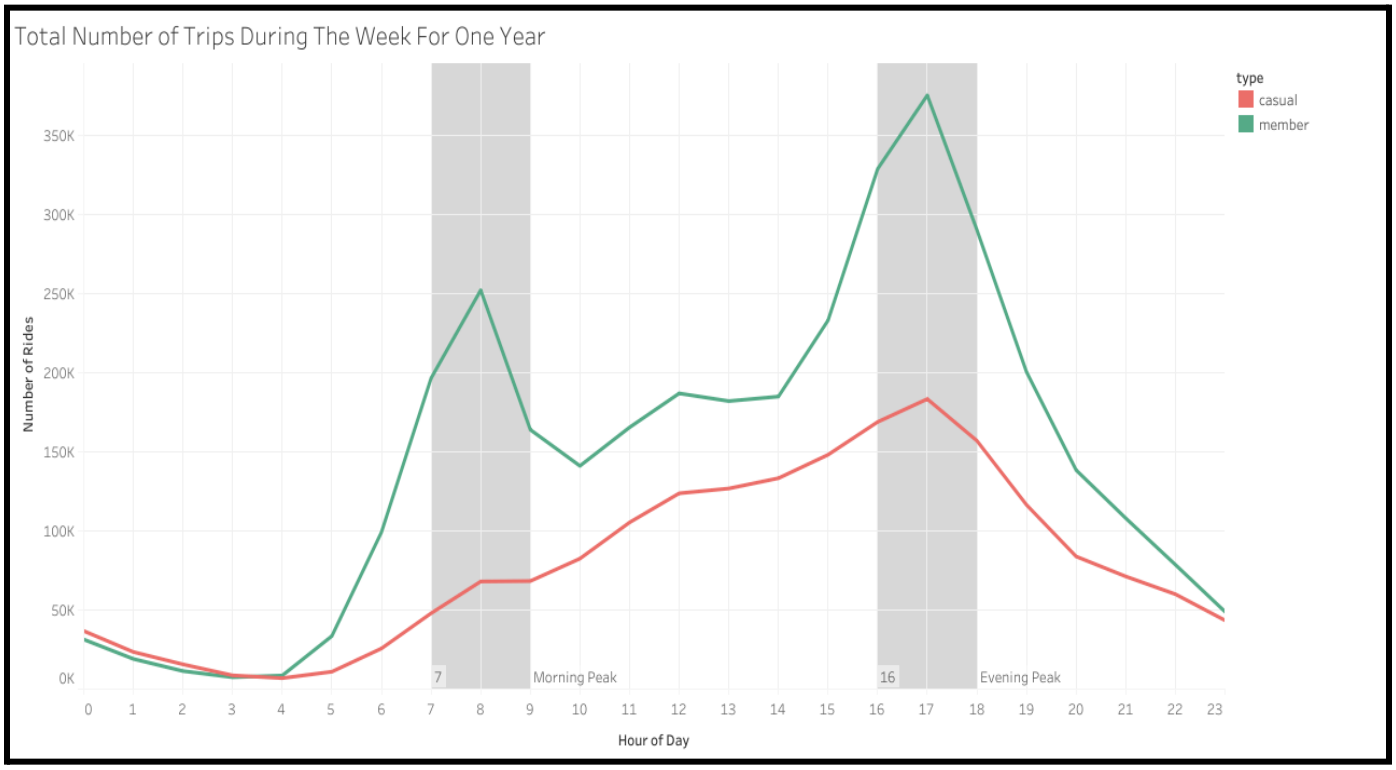
Find the volume of trips per hour and trips per day for casuals and members we plotted on the same axis in order to determine any behaviour differences between the two rider types.

Findings:

<u>Hourly Differences</u>

The line graph for annual members shows a distinct “M-shape” where rides peak during the morning hours of the business day and the afternoon hours of the business day. This indicates that annual members are mostly using their memberships for utility purposes in order to commute to and from work whether it be from a nearby train station or their homes.

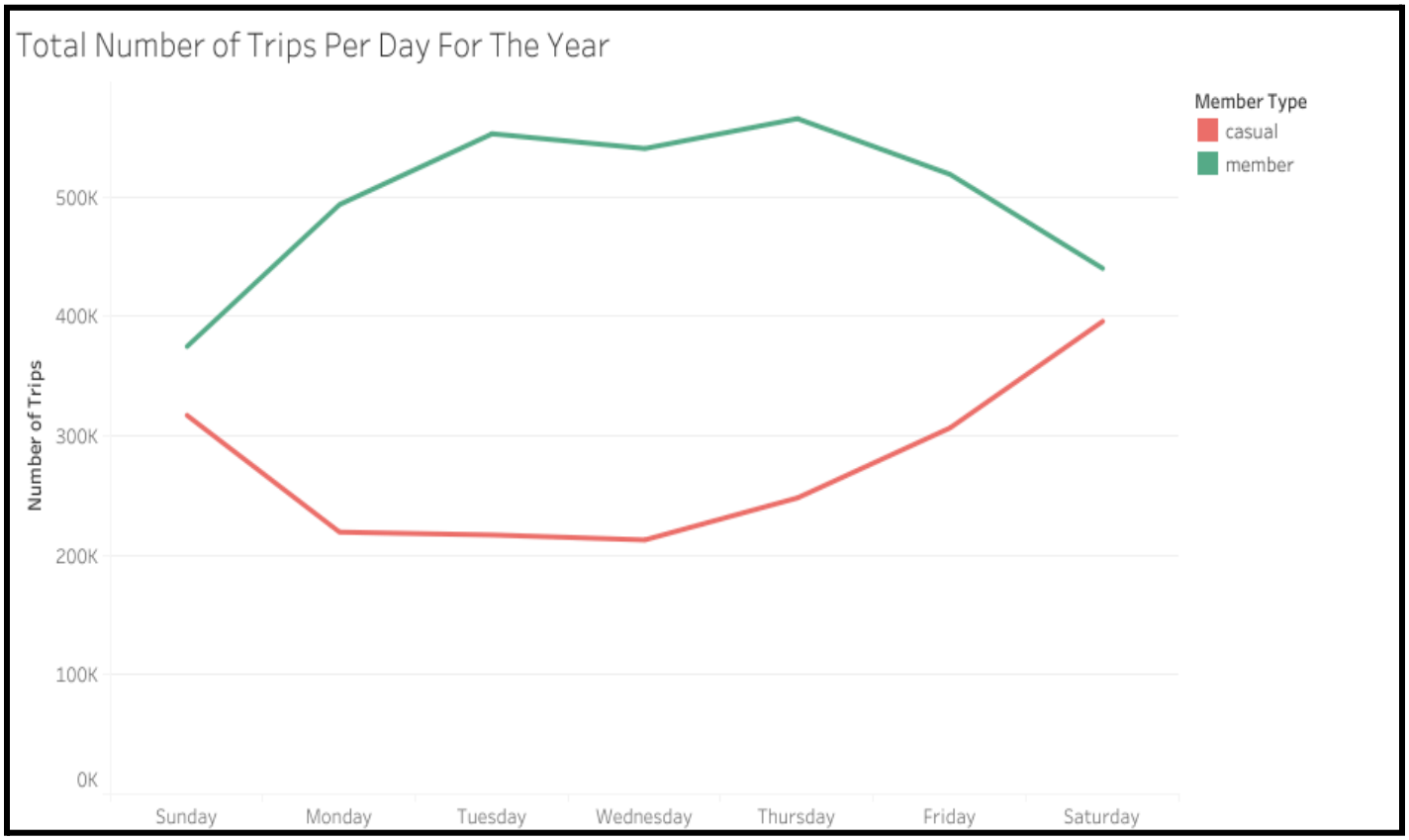
The line graph for casual members does not have this trend and instead exhibits a growing use of the bike services throughout the day till about 17h00 when usage starts to dramatically dip down again. The most likely reason for this trend is that casuals are more likely to use the bike service for leisure purposes as the rides increase throughout the day when it is most convenient for leisure activities.



Daily Differences

The line graph for annual members shows a higher usage rate than casual members throughout the week. Most trips are taken throughout the week with usage rates tapering off during the weekend at both ends on the graph. This is another signal indicating that annual members are utilizing the service to commute for work purposes.

The line for casual members essentially tells the exact opposite story with usage rates of the bike services seeing large spikes on the weekend once again indicating that casual members are utilizing the bikes on the weekend for leisure purposes.



Since we have been requested to convert more casual members to the annual memberships it would not make sense to target the entire population of casual members as the majority of them are not utilizing the service in the same manner. The best solution would be to find casuals who are exhibiting the same behaviour as annual members.

So the first characteristics we looked for in casual riders to target were the ones who shared similar travel times as members.

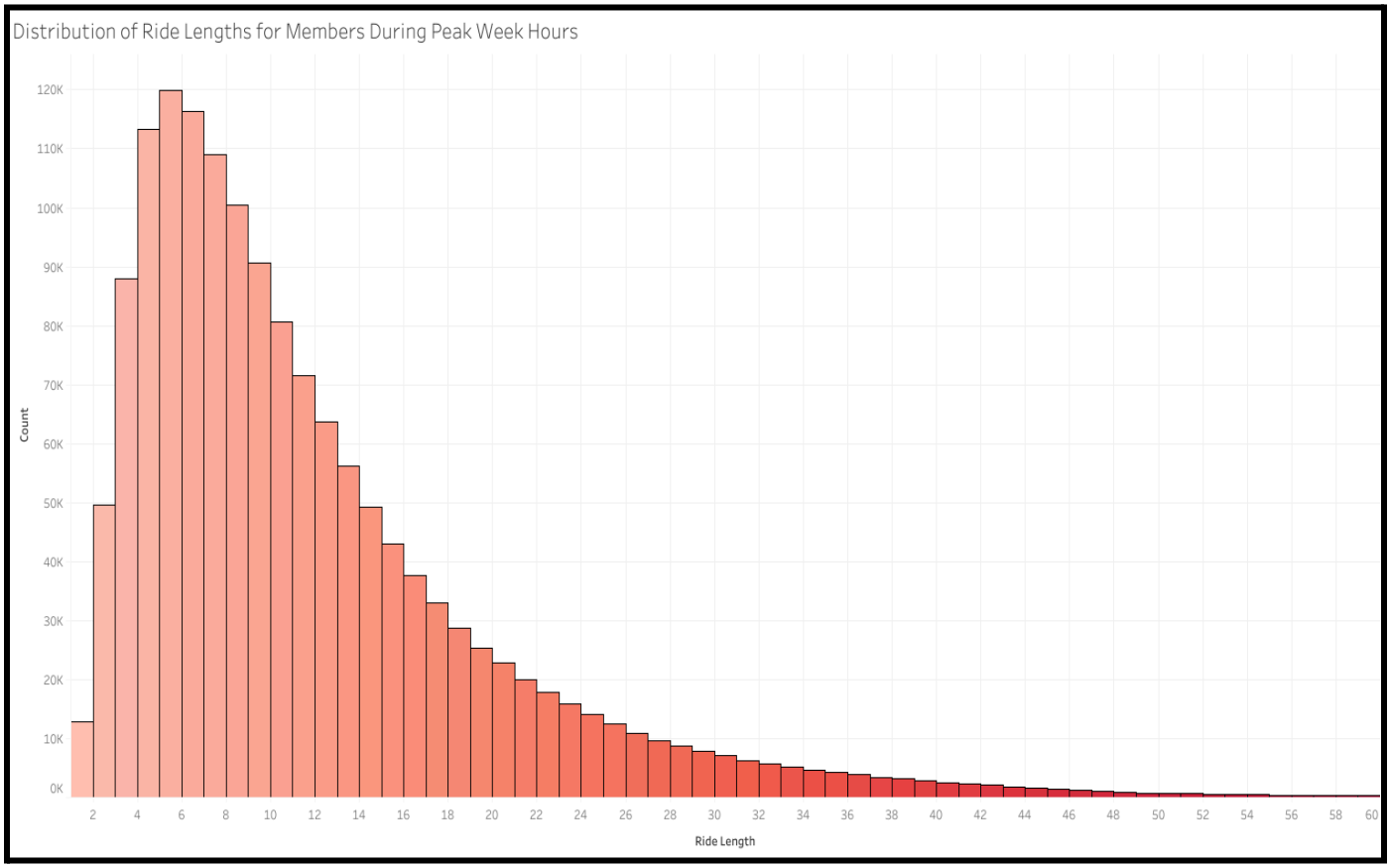
Member Ride Lengths

Analysis:

Understand the travel times of members to see what to look for in casual members when looking to target the ride casuals with digital media campaigns.

Findings:

The histogram of member travel times is majorly skewed to the right. The distribution was limited to 60 minutes in order to prevent outliers from affecting the mean and standard deviations of the travel times. We also only considered weekdays as these were the days that displayed the highest member usage rates. The mean was calculated to 11 minutes and the standard deviation to 8.56 minutes, this will help set a range of 2.54 minutes and 19.56 to look for in casual members who fit the characteristics of members. The confidence interval was only calculated with one standard deviation to ensure that casual fitted the characteristics of members.



We now knew the range of ride lengths to look for in casual members to target that being between 2.54 and 19.56.

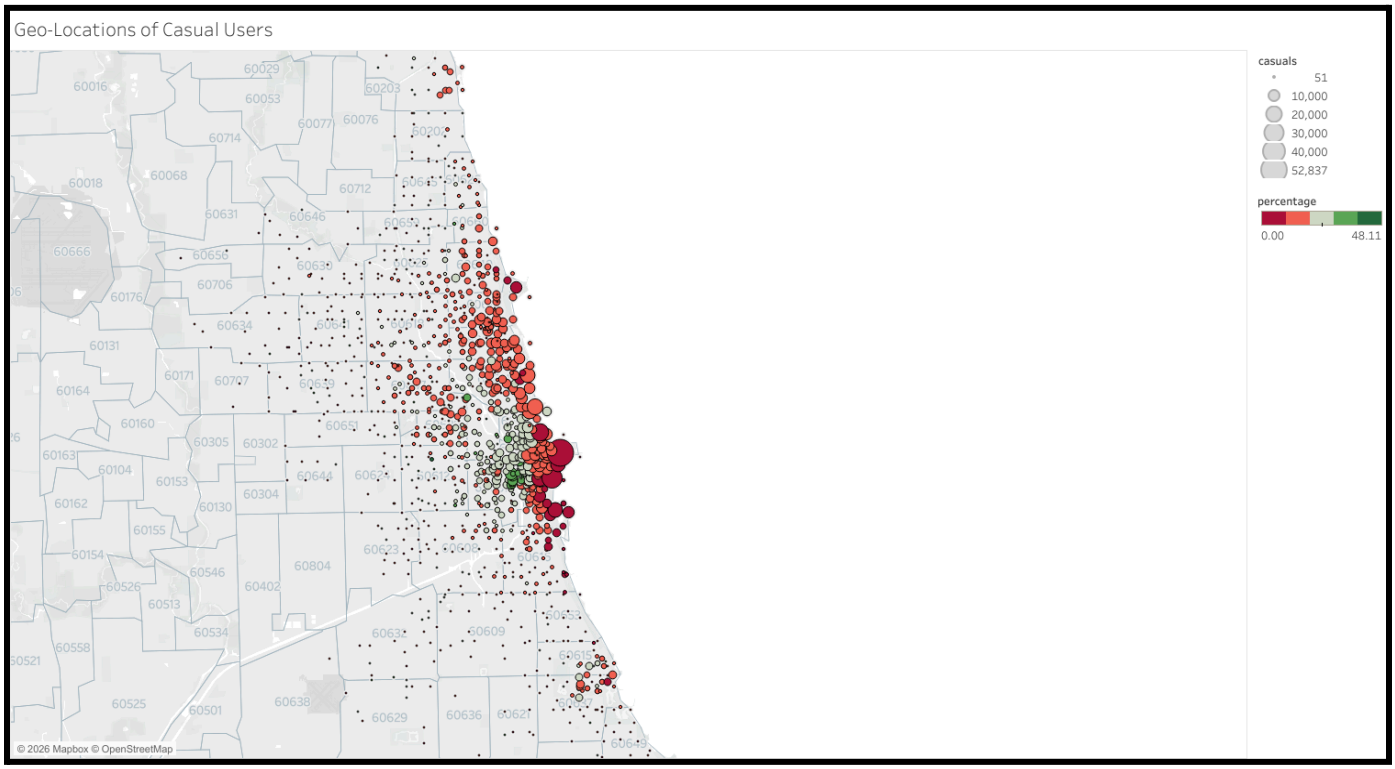
Geo-Location of Casuals

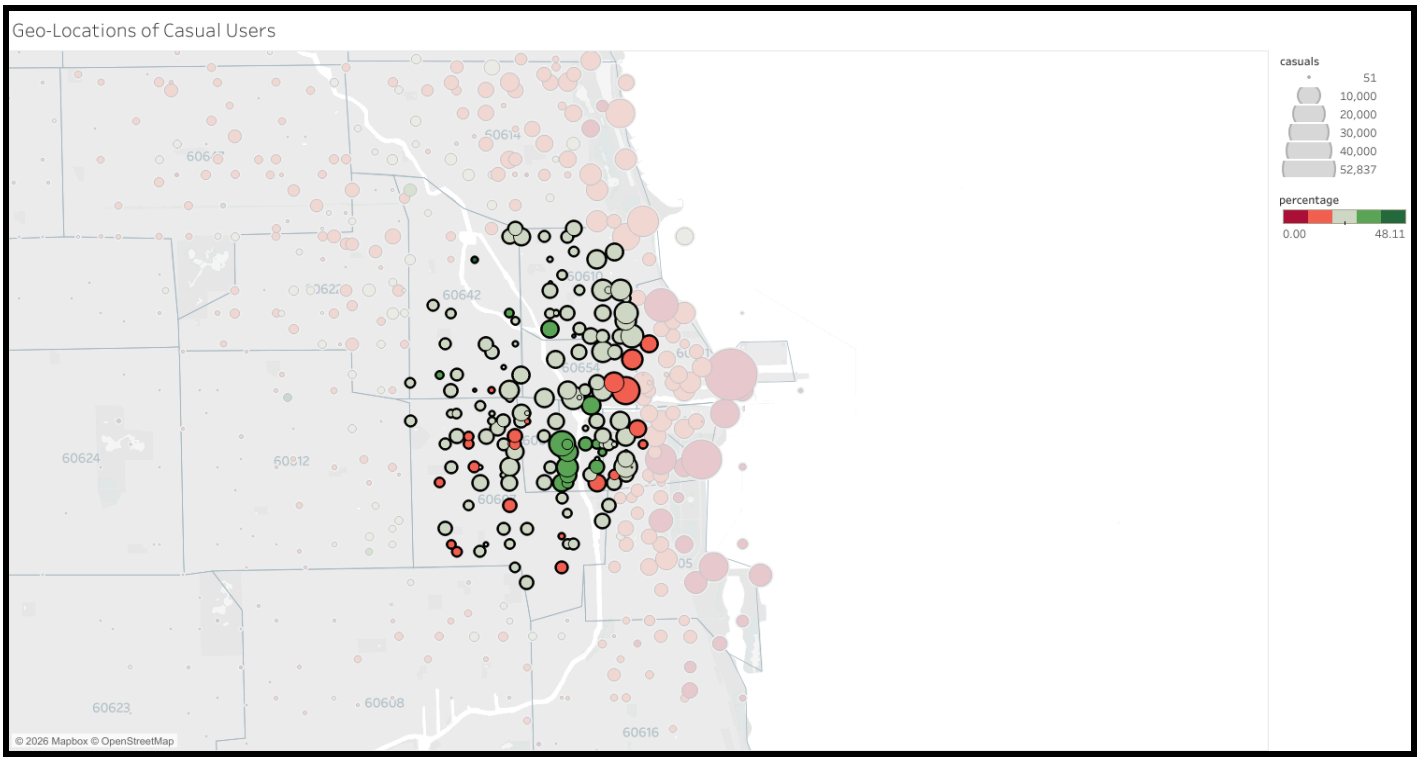
Analysis:

Since we did not have the PII of users we could not target the exact users we wanted to. With this constraint we had to find areas where casuals were demonstrating member-like behaviours. We did this by finding the geo locations of areas where casuals acted most like members. Therefore we marked the locations of areas on a map of Chicago. Each with a size of 110 square meters. With colours red to green indicating whether an area's casuals demonstrate behaviours like members.

Findings:

Most of the areas had low percentage rates of casuals who exhibited the same behaviours as members. However there appeared to be an island of green, between the sea of red. In this specific area casuals displayed the most member-like behaviour out of the rest of Chicago, with 25% - 50% of casuals displaying member-like behaviour.





Recommendations

- Geo-Located Social Media Advertisements

  Create social media adverts and set the adverts to target the zip code areas where casuals are acting like members. This will ensure that the adverts are being placed in front of the right people and will save the company money as the adverts are in specific areas where people will be interested instead of advertising to the whole of Chicago which would cost a lot more to reach the same amount of valid customers.

- Physical Advertisements in the Green Island

  Saturate the area of zip codes that form the Green Island of casuals who display member characteristics, with physical adverts such as billboards and pamphlets. Placing the adverts in areas where working people who use the bikes most likely pass by and spend their time during the work week, such as office blocks, train stations and coffee shops.

- Transit Partnerships

  Many people who travel in Chicago use trains/subways to get into the city and then bike from these stations to their final destination using the rentals to travel. Cyclistic can partner with these transit hubs within the specified Green Island to promote Cyclistic bike memberships within these hubs